

Guitar String Vibration Analysis Using Smartphone Accelerometer Data and FFT

Prepared by : Arezoo Shirmohammadi

September 2026

1. Introduction

The vibration of a guitar string is a simple physical system that demonstrates periodic motion, frequency, and harmonic behavior. When a string is plucked, it vibrates at a fundamental frequency that determines its musical pitch. Higher-frequency components, known as harmonics, may also be present.

In this project, a smartphone accelerometer was used to investigate the vibrations of guitar strings. Acceleration data were recorded using the phyphox application and subsequently analyzed in MATLAB.

The Fast Fourier Transform (FFT) was used to identify the dominant frequencies in the measured signals. The measured frequencies were compared with the reference frequencies of the corresponding guitar strings. Repeated measurements were also used to evaluate the consistency of the results.

The main objectives were to:

- measure guitar-string vibrations using a smartphone;
- determine their fundamental frequencies using FFT;
- compare measured and reference frequencies;

- evaluate measurement accuracy and repeatability;
- investigate the harmonic structure of the E2 string.

2. Experimental Setup

A smartphone equipped with an accelerometer was placed on the guitar body to detect mechanical vibrations produced by the strings. The acceleration signal along the X-axis was recorded using phyphox.

Four strings were investigated, with five repeated measurements for each string.

String	Theoretical fundamental frequency
E2	82.41 Hz
A2	110.00 Hz
D3	146.83 Hz
G3	196.00 Hz

The recorded data were exported as Excel files and processed in MATLAB.

The sampling frequency was approximately 470 Hz, giving a Nyquist frequency of approximately 235 Hz. Therefore, the four investigated strings were within the measurable frequency range.

3. Data Analysis

The recorded acceleration signals were processed in MATLAB before frequency analysis.

The main processing procedure was:

Acceleration data → preprocessing → vibration detection → 4-second segment → Hann window → FFT → frequency extraction

The average value of the acceleration signal was first removed. The beginning of the main vibration was then detected automatically, and a 4-second segment was selected for analysis.

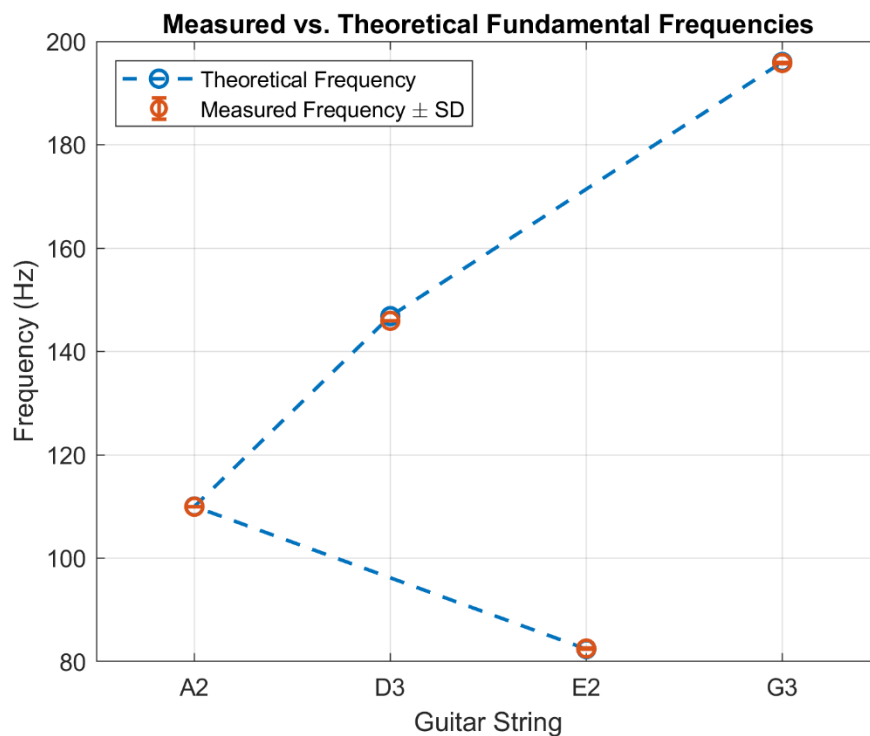
A Hann window was applied to reduce spectral leakage. Finally, the FFT was used to convert the signal from the time domain into the frequency domain.

The fundamental frequency was identified as the strongest frequency peak near the expected reference frequency.

Detailed MATLAB scripts and processing files are available in the GitHub repository.

4. Fundamental Frequency Results

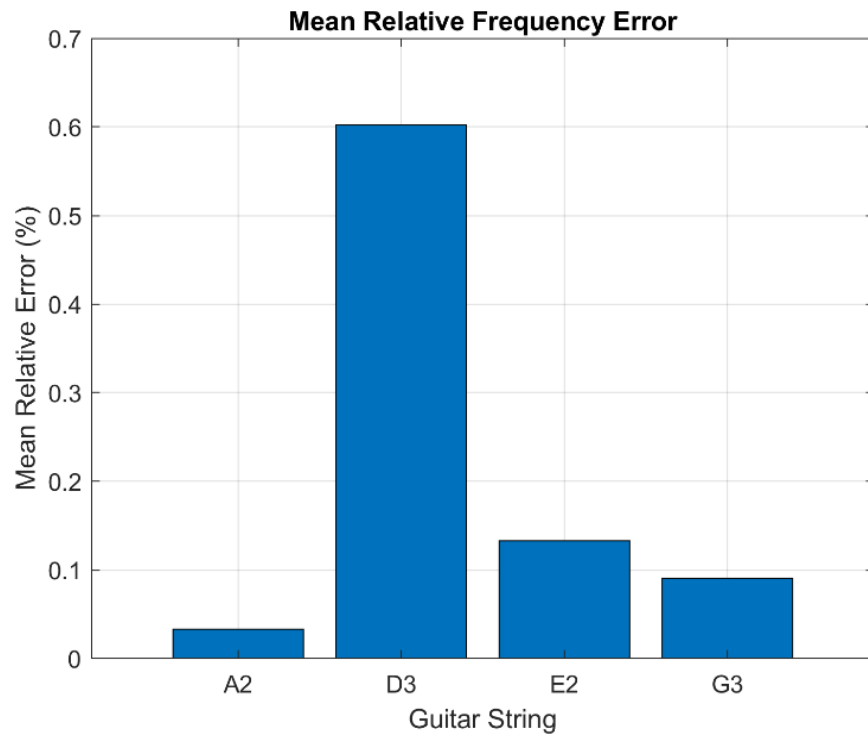
The measured mean frequencies were close to the reference frequencies for all four strings.



The mean measured frequencies were:

String	Reference (Hz)	Mean Measured (Hz)	Mean Relative Error
E2	82.41	82.52	0.133%
A2	110.00	109.96	0.033%
D3	146.83	145.95	0.602%

G3	196.00	195.82	0.09%
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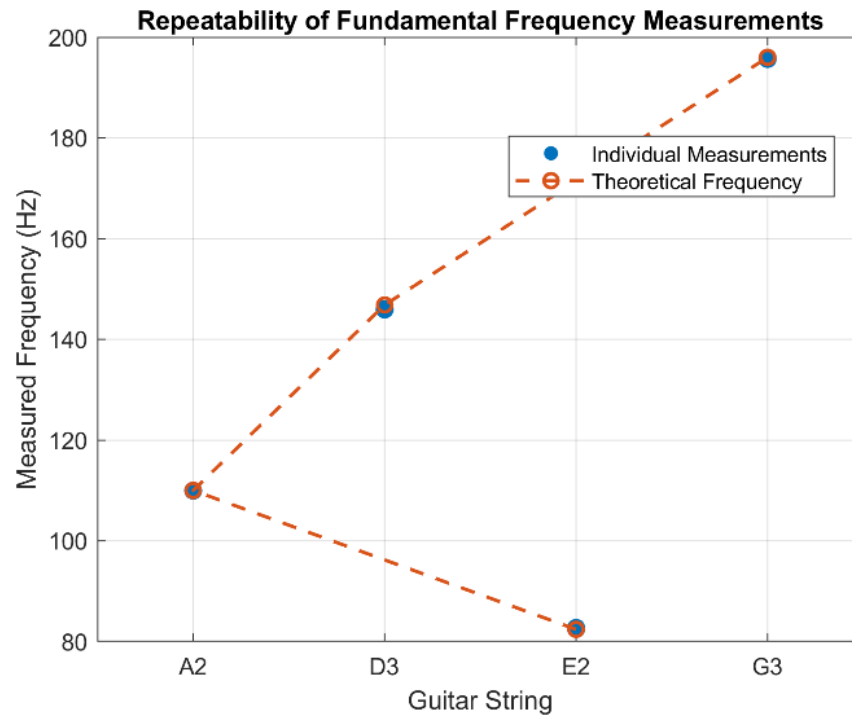


The relative error was below 1% for all four strings. D3 showed the largest deviation, with a mean error of approximately 0.60%.

However, the D3 measurements were extremely close to one another. This indicates that the measurement was highly repeatable, despite having a small systematic difference from the reference frequency.

5. Repeatability

Five measurements were performed for each string to evaluate the consistency of the frequency measurements.



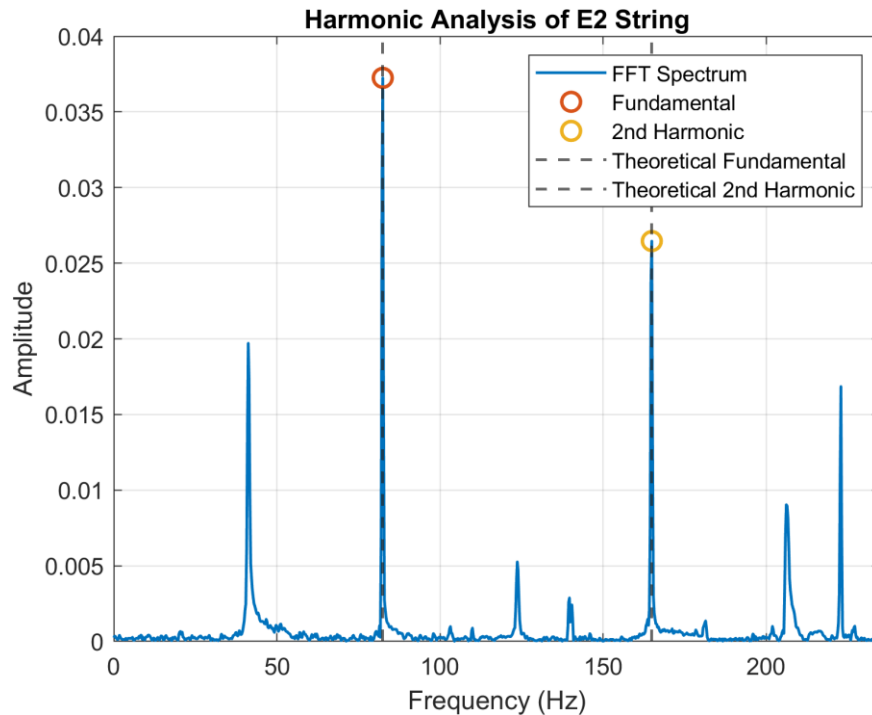
The repeated measurements showed generally small variations. In particular, A2 and D3 showed very high repeatability.

The results demonstrate an important distinction between accuracy and repeatability. Accuracy describes how close the measurement is to the reference value, whereas repeatability describes how close repeated measurements are to one another.

The D3 measurements provide a clear example: they were highly consistent but consistently lower than the reference frequency.

6. Harmonic Analysis

The E2 string was selected for a more detailed frequency-domain analysis.



The fundamental frequency was approximately 82.5 Hz, while a second harmonic was detected near 164.9 Hz.

The second harmonic is approximately twice the fundamental frequency, which is consistent with the expected harmonic structure of a vibrating string.

The second harmonic was also detected in all five E2 measurements. Its frequency location was highly consistent, although its measured amplitude varied considerably between measurements.

This suggests that frequency measurements were more stable than amplitude measurements under the experimental conditions used in this study.

7. Discussion

The results show that a smartphone accelerometer can be used as a simple and low-cost tool for investigating guitar-string vibrations.

The fundamental frequencies obtained from the four strings were close to their reference values, with errors below 1%. The repeated measurements also demonstrated good frequency repeatability.

The small systematic offset observed for D3 may be related to experimental factors such as string tension, tuning, sensor placement, or the physical interaction between the guitar body and the smartphone.

The harmonic analysis of E2 further demonstrated that the FFT spectrum can reveal not only the fundamental frequency but also higher-frequency components of the vibration.

8. Limitations

The experiment has several limitations. The smartphone accelerometer is not a dedicated vibration sensor, and the measured amplitude can be strongly affected by sensor placement and the mechanical coupling between the phone and guitar.

In addition, the sampling frequency limited the maximum directly measurable frequency to approximately 235 Hz. As a result, only the four lower guitar strings were analyzed in the main frequency study.

9. Conclusion

This project demonstrated the use of a smartphone accelerometer and FFT analysis to investigate guitar-string vibrations.

The measured fundamental frequencies were generally close to the expected reference values, with relative errors below 1%. Repeated measurements showed good frequency repeatability, although accuracy and repeatability were not always identical.

The E2 analysis also revealed a second harmonic near twice the fundamental frequency, demonstrating the ability of the measurement system to capture basic harmonic features of guitar-string vibration.

Overall, the results show that smartphone-based sensing combined with MATLAB signal processing can provide a useful and accessible approach for experimental investigation of mechanical vibrations.